



ZAALIMA DEVELOPMENT INTERNSHIP

WEEK 2 REPORT

Name: Pooja M

Project: Customer Churn Prediction & Lifetime Value (LTV) Engine

Project Objective

Develop machine learning models to predict customer churn and identify important factors affecting retention.

Feature Engineering

Created Average Monthly Spend and Long-Term Customer features to improve model performance.

Data Preprocessing

Encoded categorical variables, split training/testing datasets, and prepared data for modeling.

Model Development

Implemented Logistic Regression and Random Forest algorithms for churn prediction.

Model Evaluation

Measured Accuracy, Precision, Recall, and F1-Score to evaluate model performance.

Feature Importance

Identified Contract Type, Tenure, Monthly Charges, and Total Charges as major factors.

Business Impact

Predictions help companies target high-risk customers and improve retention strategies.

Outcome

Successfully built and evaluated predictive models with actionable insights.